UNIT 1 FRQ

Include correctly labeled diagrams, if useful or required, in explaining your answers. A correctly labeled diagram must have all axes and curves clearly labeled and must show directional changes. If the question prompts you to "Calculate," you must show how you arrived at your final answer.

Assume both Atlantis and Zanadu produce helmets and baseballs. Using equal amounts of resources, Atlantis can produce 100 helmets or 200 baseballs, whereas Zanadu can produce 100 helmets or 400 baseballs.

(a) Assume both Atlantis and Zanadu experience constant opportunity costs in producing helmets and baseballs. Draw a correctly labeled graph illustrating the production possibilities curves (PPCs) for Atlantis and Zanadu, showing helmets on the horizontal axis and baseballs on the vertical axis. Plot the numerical values provided above on your graph.

(b) Calculate the opportunity cost of one helmet for Atlantis. Show your work.

(c) Which country has an absolute advantage in the production of baseballs?

(d) Which country has a comparative advantage in the production of baseballs? Explain.

(e) If Atlantis and Zanadu specialize based on comparative advantage and trade, would they be able to gain from trade if the terms of trade are 1 helmet for 3 baseballs? Explain.

1. Respond to all parts of the question.

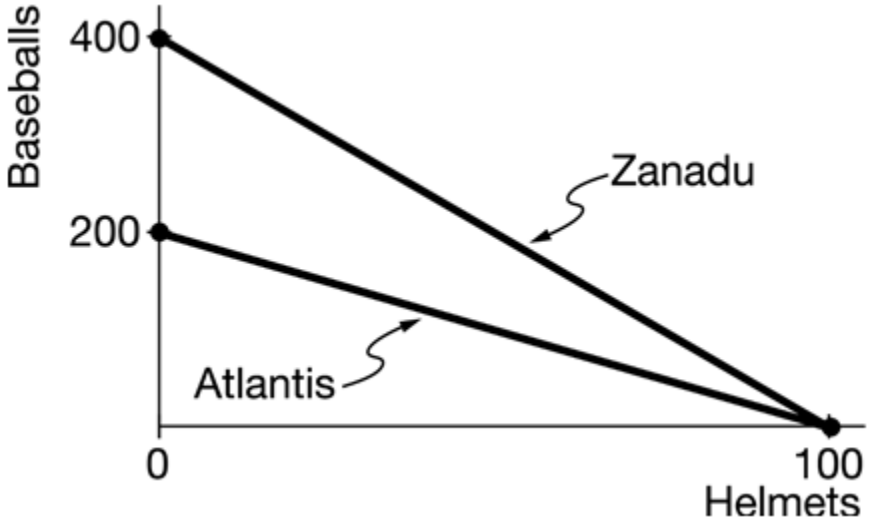
Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a correctly labeled graph of the production possibilities curves for Atlantis and Zanadu with helmets on the horizontal axis and baseballs on the vertical axis and using the numerical values provided (see graph below).

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Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response correctly calculates the opportunity cost of one helmet for Atlantis as 2 baseballs and shows in the work that the $\text{opportunity cost} = 200 \text{ baseballs}/100 \text{ helmets} = 2 \text{ baseballs per helmet}$

Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response states that Zanadu has an absolute advantage in the production of baseballs.

Part (d)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response indicates that Zanadu has a comparative advantage in the production of baseballs and explains that Zanadu has a lower opportunity cost of producing a baseball ($1/4$ of a helmet) than Atlantis's opportunity cost of producing a

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baseball ($1/2$ of a helmet).

Part (e)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response correctly answers with "yes" and explains that the terms of trade fall between the opportunity costs of producing helmets in the two countries (i.e., between 2 baseballs in Atlantis and 4 baseballs in Zanadu).

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*I*nclude correctly labeled diagrams, if useful or required, in explaining your answers. A correctly labeled diagram must have all axes and curves clearly labeled and must show directional changes. If the question prompts you to “Calculate,” you must show how you arrived at your final answer.

The data provided below describe an ice-cream store’s daily production possibilities for milkshakes and sundaes.

Milkshakes	0	20	40	60	80	100	120
Sundaes	210	200	180	150	110	60	0

(a) Draw a correctly labeled graph of the store’s production possibilities curve (PPC) with milkshakes on the horizontal axis and sundaes on the vertical axis and label the endpoints using the numbers provided above.

(b) Is the opportunity cost of producing milkshakes increasing, decreasing, or constant? Explain using numbers in the table.

(c) The store is currently producing 80 milkshakes and 110 sundaes. Calculate the opportunity cost of increasing milkshake production from 80 to 100 milkshakes. Show your work.

(d) Would it be efficient for the store to produce 80 milkshakes and 150 sundaes? Explain using numbers in the table.

(e) Suppose the ice-cream store purchases a new machine that increases the output of milkshakes and sundaes. Show the effect of the change on the store’s PPC on your graph in part (a).

2. Respond to all parts of the question.

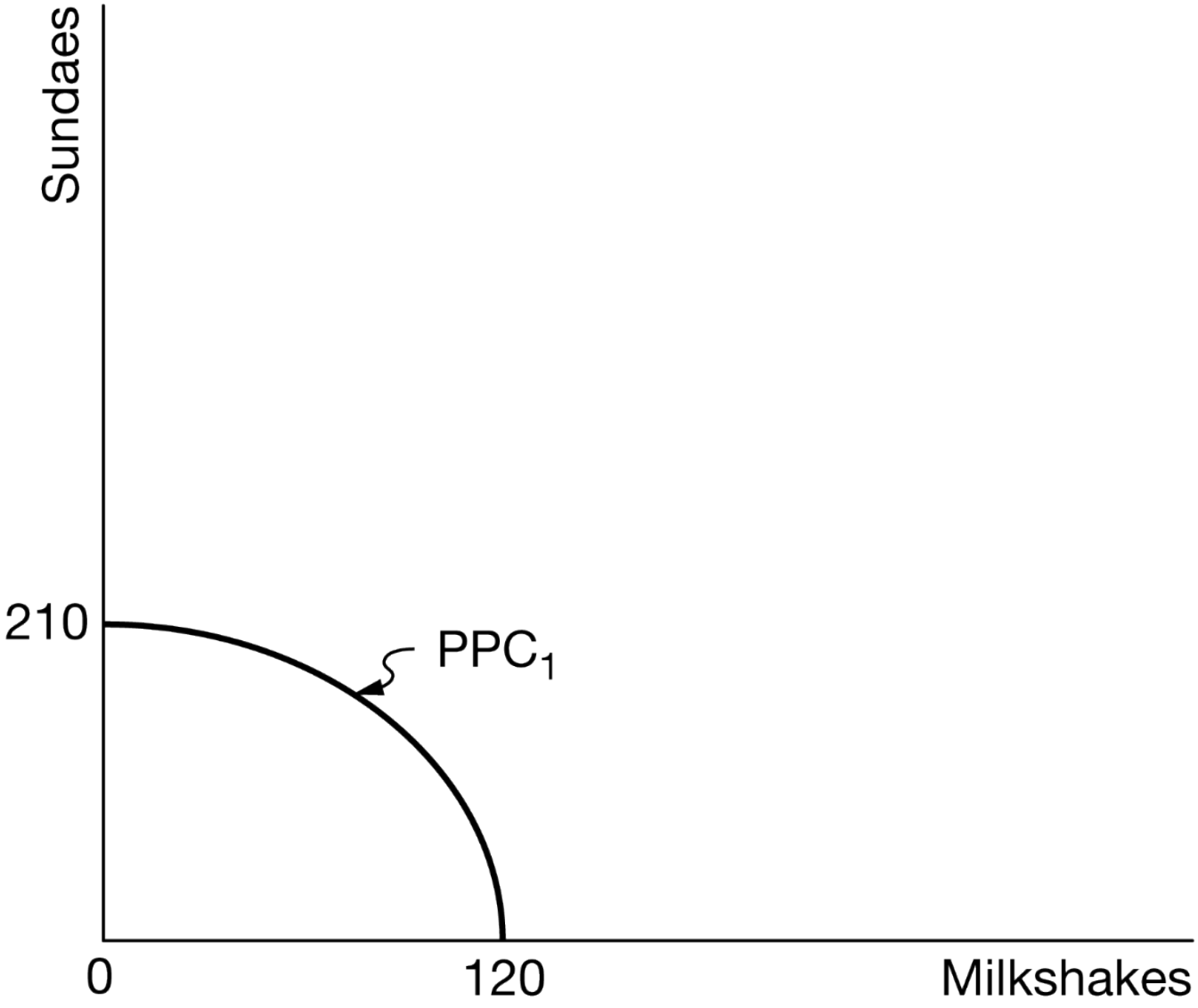
Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1 ✓

The response draws a correctly labeled graph of the production possibilities curve (PPC) for the store with milkshakes on the horizontal axis and sundaes on the vertical axis and the quantities noted.



Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1 ✓

The response correctly states that the opportunity cost of producing milkshakes is increasing and explains that as the production of milkshakes increases by the same amount of 20 milkshakes, the production of sundaes decreases at an increasing rate. The first 20 milkshakes cost 10 sundaes. The last 20 milkshakes cost 60 sundaes. Therefore, the opportunity cost of producing milkshakes increases as more milkshakes are produced.

Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response calculates the opportunity cost as 50 sundaes and shows the work $(110 - 60 = 50)$.

Part (d)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response states that the store cannot produce 80 milkshakes and 150 sundaes using its available resources and explains that it can produce 80 milkshakes and 110 sundaes or 60 milkshakes and 150 sundaes but not a larger combination of both products.

Part (e)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

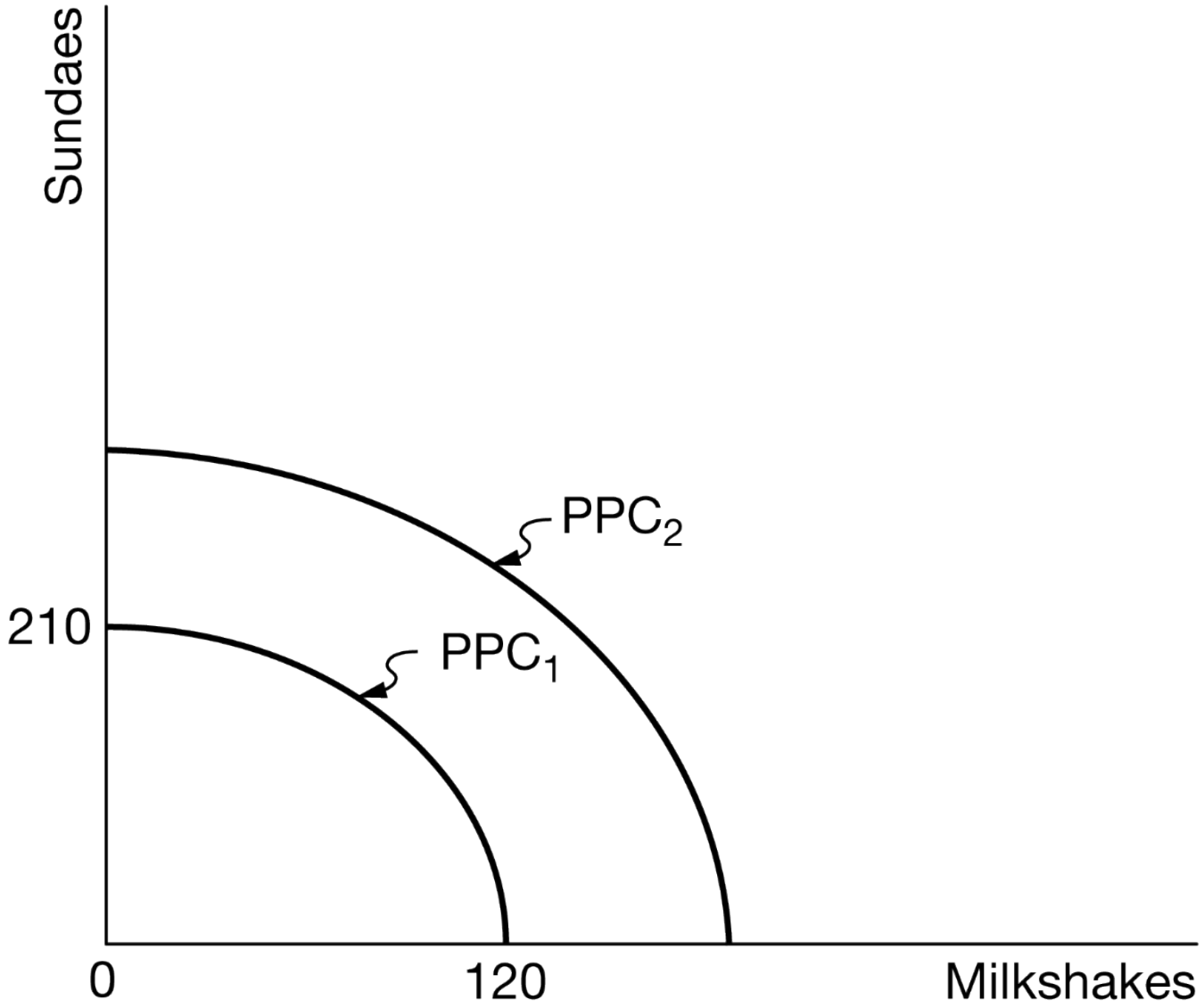
This point is dependent on drawing a correctly labeled graph of the store's (PPC in part (a).

0	1 ✓

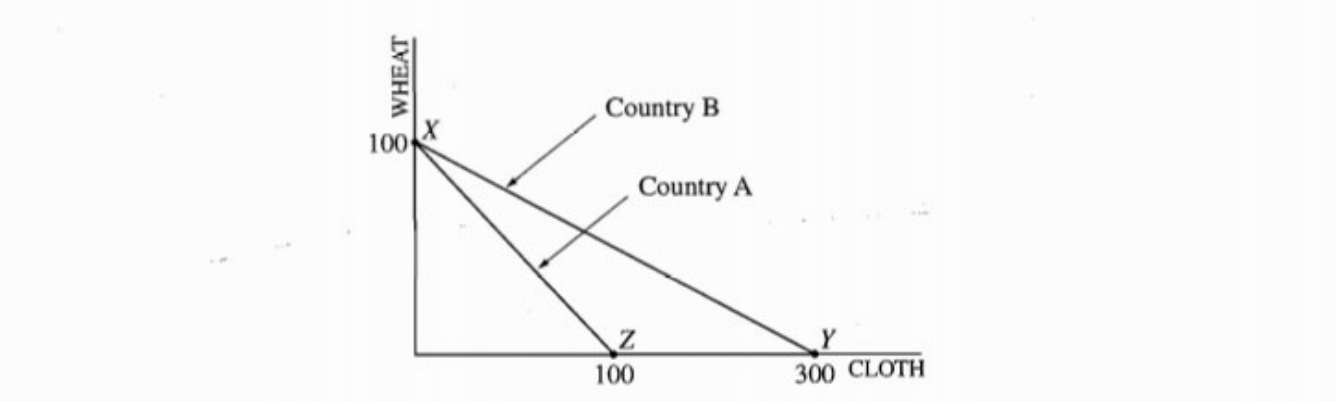
The response correctly shows the effect of the change with an outward shift of the store's PPC

(PPC_2) that shows increasing quantities of the two products.

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Assume that Countries A and B have equal amounts of resources and identical technologies. Country A can produce 100 bushels of wheat or 100 yards of cloth or any combination, as shown by the line *XZ* in the figure above. Country B can produce 100 bushels of wheat or 300 yards of cloth or any combination, as shown by the line *XY* in the figure above.

a. Which country has an absolute advantage in the production of wheat and which has an absolute advantage in the production of cloth? Explain how you determined your answer.

b. Which country has a comparative advantage in the production of wheat and which has a comparative advantage in the production of cloth? Explain how you determined your answer.

c. With specialization and trade, which country will import wheat? Explain why.

d. Assume that the two countries trade, and that one bushel of wheat is exchanged for two yards of cloth. Explain why the country that imports wheat will gain from trade.

3. Respond to all parts of the question.

Part A

1 Point

- **1 -** for Wheat: neither country has an absolute advantage, identical outputs (with the same resource) Cloth: Country B, as it produces more output (with the same resources)

0	1 ✓

The student earns all of the following points:

- **1 -** for Wheat: neither country has an absolute advantage, identical outputs (with the same resource) Cloth: Country B, as it produces more output (with the same resources)

Part B

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2 Points

- **1 -** for Country A has comparative advantage in wheat; Country B in cloth
- **1 -** for shown opportunity costs; Country A give up 1 cloth for 1 wheat; Country B gives up 3 cloth for 1 wheat; relatively more expensive for Country B to produce wheat

0	1	2 ✓

The student earns all of the following points:

- **1 -** for Country A has comparative advantage in wheat; Country B in cloth
- **1 -** for shown opportunity costs; Country A give up 1 cloth for 1 wheat; Country B gives up 3 cloth for 1 wheat; relatively more expensive for Country B to produce wheat

Part C

1 Point

- **1 -** for Country B will import wheat because it has comparative advantage in cloth (and will specialize in cloth production, getting its wheat more cheaply via trade) Key point is comparative advantage

0	1 ✓

The student earns all of the following points:

- **1 -** for Country B will import wheat because it has comparative advantage in cloth (and will specialize in cloth production, getting its wheat more cheaply via trade) Key point is comparative advantage

Part D

1 Point

- **1 -** for country B, via trade a unit of wheat only costs 2 units of cloth, while by domestic production a unit of wheat would cost 3 units of cloth. Also the student may explain that trade increases B's consumption possibilities by making it possibly to consumer more of both goods

0	1 ✓

The student earns all of the following points:

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- **1 -** for country B, via trade a unit of wheat only costs 2 units of cloth, while by domestic production a unit of wheat would cost 3 units of cloth. Also the student may explain that trade increases B's consumption possibilities by making it possibly to consumer more of both goods